

Math Special Interest Group

May 13, 2026

Agenda

- Discussion on oneMath spec – John Melonakos (Intel)
- Opens
- Updates from last meeting – Sarah Knepper
 - oneMath open source project and specification

SYCL buffers/accessors

- SYCL specification is potentially planning to make buffers/accessors optional.
- Any feedback to share?
- Any value in returning events from SYCL buffer APIs in oneMath? This would align the return type between USM and buffer APIs.

Proposal: Merge Math SIG into AI and Scientific Computing SIG

- AI Special Interest Group (SIG) evolved into the AI & Scientific Computing SIG, expanding its scope to include scientific frameworks and AI for Science.
 - Driving technical discussions and knowledge sharing on AI frameworks and Scientific Computing, with UXL software serving as the foundation for scientific frameworks and applications.
 - Acting as a bridge between scientific R&D communities and developers to foster collaboration and innovation
 - Host regular meetings to share community proposals, case studies, technical advancements, challenges, and contributions, and provide feedback and requirements to guide UXL component development
 - Identifying common patterns and requirements in scientific computing and aligning them with AI Surrogate Models & Frameworks

oneMath Open Source Project and Specification

- In progress: OpenBLAS backend implementation ([#722](#))
- Use dynamic link for oneMKL, add MKLCPU_THREAD_OMP ([#718](#))
- [cmake] remove the latest lib directories from MKLConfig ([#719](#))
- Fix failing ArmPL SCAL tests ([#723](#))
- Numerous other improvements

Recent updates to oneMath specification:

- [BLAS] Deprecate 'gemmt' and rename it to 'gemmtr' ([#667](#))

Resources

- Unified Acceleration Foundation: <https://uxlfoundation.org/>
- GitHub for UXL Foundation: <https://github.com/uxlfoundation/foundation>
- Latest release of oneMath Spec (currently 1.4): <https://oneapi-spec.uxlfoundation.org/specifications/oneapi/latest/elements/onemath/source/>
- GitHub for oneAPI Spec: <https://github.com/uxlfoundation/oneAPI-spec>
- GitHub for oneMath open source project (currently BLAS, RNG, LAPACK, DFT, and sparse BLAS domains): <https://github.com/uxlfoundation/oneMath>
- [UXL Foundation Slack](#) [#sig-math](#) [#onemath](#) channels